

Experiment 9

Question

Perform Rotation of an Image in Clockwise and Counter-Clockwise Directions.

Aim

To read an image using Python and OpenCV and rotate it in both clockwise and counter-clockwise directions.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Rotate the image **90° clockwise**.
6. Rotate the image **90° counter-clockwise**.
7. Display both rotated images.
8. Save the rotated images.

Code

```
import cv2

# Read image

image = cv2.imread("penguin.jpg")

# Check whether image is loaded

if image is None:

    print("Image not found!")

else:

    # Display original image

    cv2.imshow("Original Image", image)

    # Rotate clockwise (90°)

    clockwise = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

    # Rotate counter-clockwise (90°)

    counter = cv2.rotate(image, cv2.ROTATE_90_COUNTERCLOCKWISE)

    # Display rotated images

    cv2.imshow("Clockwise Rotation", clockwise)

    cv2.imshow("Counter Clockwise Rotation", counter)

    # Save rotated images

    cv2.imwrite("clockwise_penguin.jpg", clockwise)

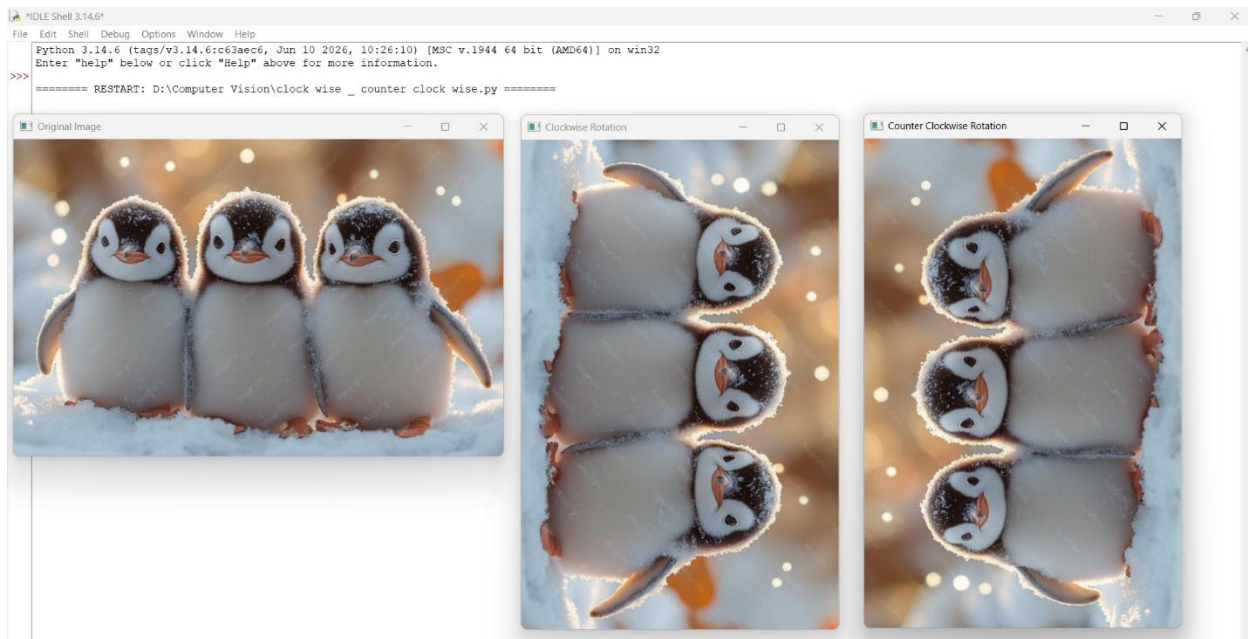
    cv2.imwrite("counter_penguin.jpg", counter)
```

Input

Input Image: penguin.jpg

Output

1. Original Image
2. Clockwise Rotated Image (90°)
3. Counter-Clockwise Rotated Image (90°)



Result: The image was successfully rotated by 90° clockwise and 90° counter-clockwise using OpenCV. The rotated images were displayed and saved successfully.